



Lecture 8

Rheology



Definition of Rheology

- **Rheo** = flow & **logy** = science so it means flow or deformation of materials
 - Rheology → is the science/physics that concerns with the flow of liquids and the deformation of solids.
 - **Fluids include both liquids and gases**
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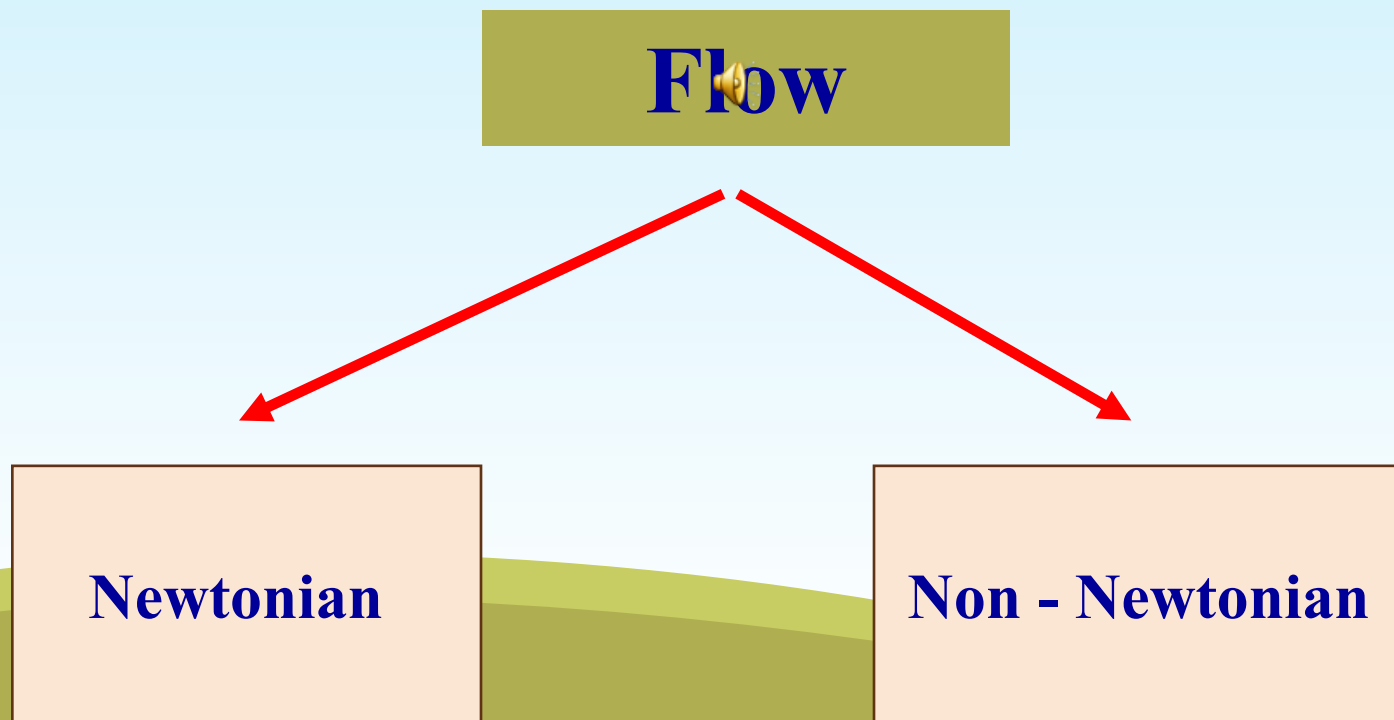
Fundamentals of Rheology

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- i. **Manufacturing of dosage forms:** Materials undergo process such as mixing, flowing through pipes, filling into the containers etc. Flow related changes influence the selection of mixing equipment.

 - ii. **Handling of drugs for administration:** The syringibility of the medicines, the pouring of the liquids from containers, extrusion of ointment from tubes, all depend on the changes in flow behavior of dosage forms.

Newton was the first to study the flow properties of liquids in quantitative terms. Liquids that obey Newton's law of flow are called as *Newtonian fluids*.

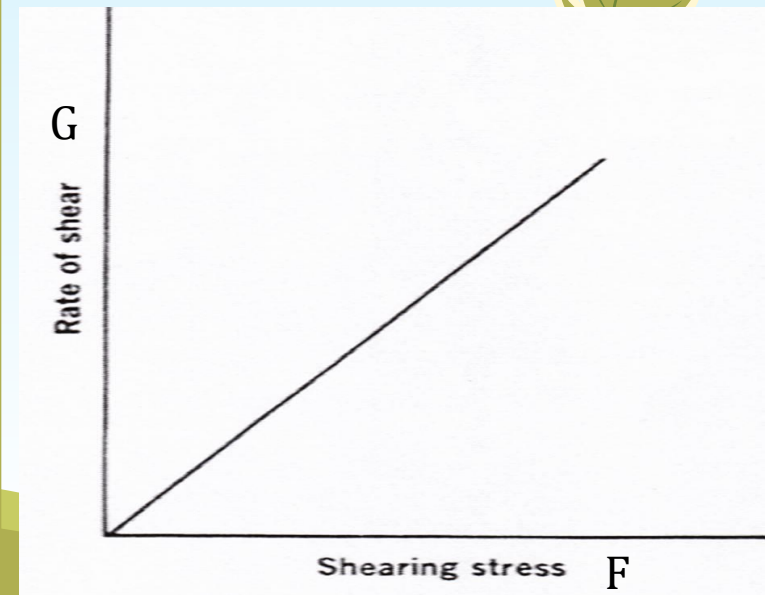
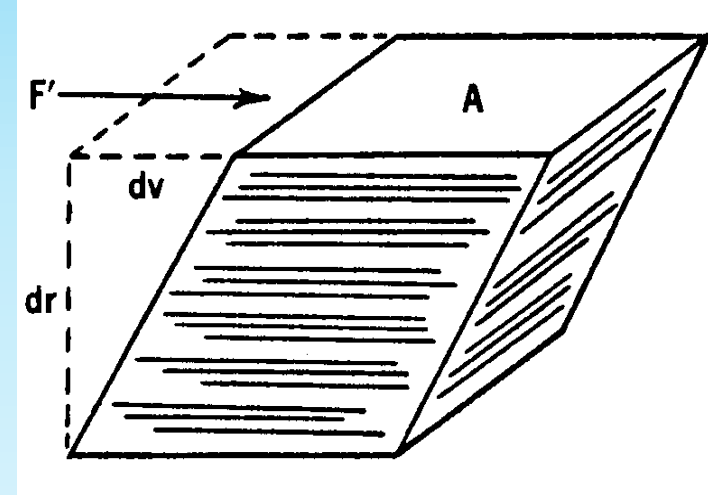
Newtonian and Non-Newtonian Flows



Newtonian Flow

- For *Newtonian fluids*.
- The force per unit area creating or producing the flow is called the **Shear Stress F** ($F = F' / A$)
- The gradient of the velocity is called the **Shear Rate G** (velocity gradient) ($G = dv / dr$)
- The rate of shear is directly proportional to the shearing stress

$$F = \eta G$$



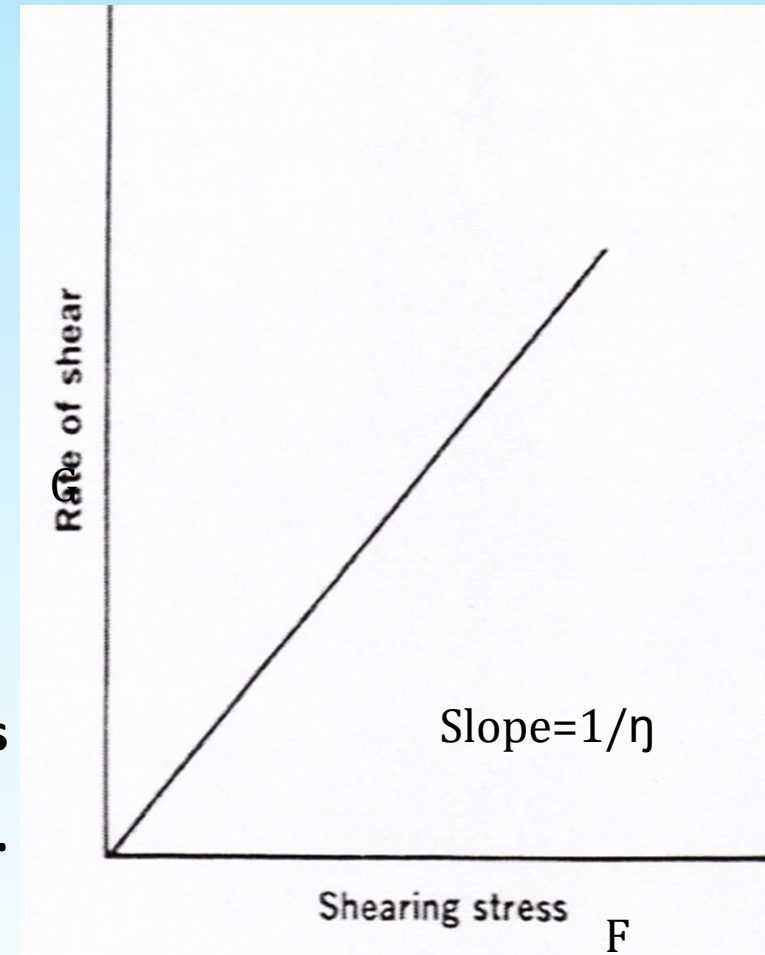
- **Viscosity** = resistance of fluid to flow
- The higher the viscosity of a liquid, the greater the force per unit area (shearing stress) required to produce a certain rate of shear.
- *Newtonian systems have **constant viscosity** independent on rate of shear ex: water , glycerin*
- it is measured in **poise**
- 1 cp = 0.01 poise
- **Fluidity (Φ)** is the reciprocal of viscosity = $1 / \eta$
- **Kinematic viscosity** = Viscosity / Density $\rightarrow \eta / \rho \rightarrow$ stoke (s) & centistoke

Rheogram (Flow curve)

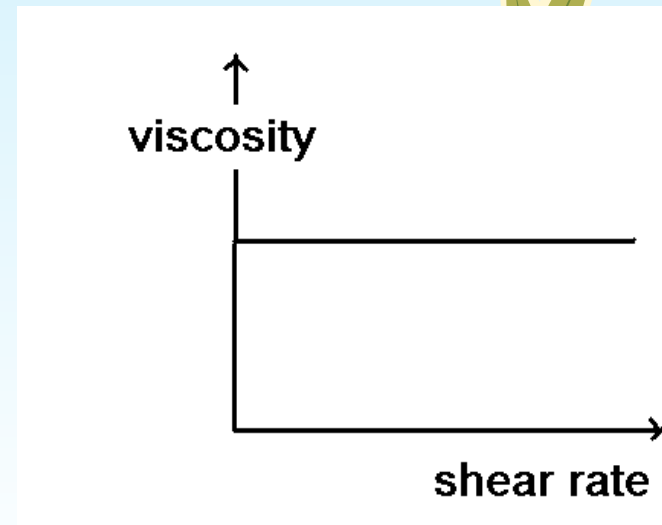
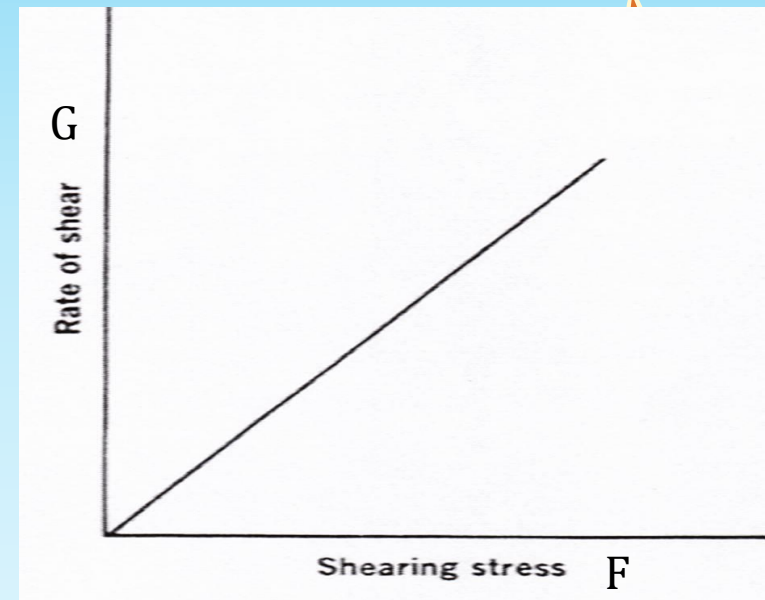
- A **Rheogram** is a plot of the relationship between the applied shear stress and the resultant shearing rate

For a Newtonian Fluid:

The Rheogram is a straight line passing through the origin: the higher the shear stress applied, the higher the obtained rate of shear.



- The slope of the rheogram is the **reciprocal** of the **absolute viscosity**
- Curve is a **straight** line passing through the **origin**.
- The slope is **constant** indicating the persistence of the value of viscosity however the applied shear stress and the proportionally obtained rate of shear.
- This is characteristic of newtonian fluids





Effect of temperature on viscosity

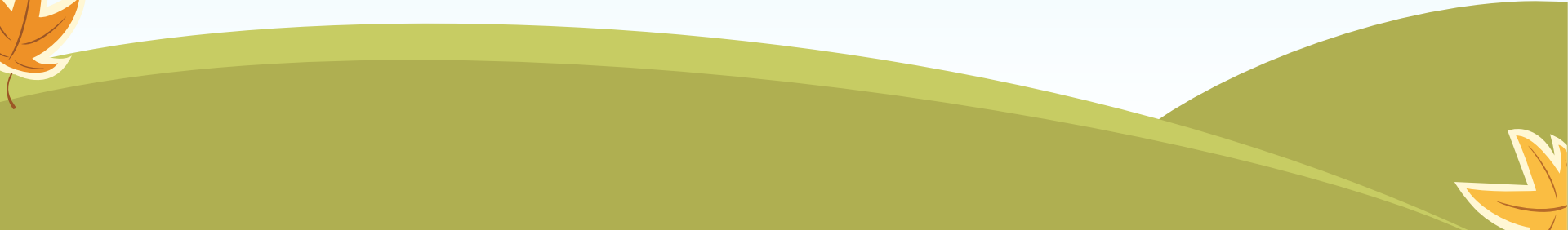
- For a Newtonian fluid, the viscosity, by definition, depends only on **temperature and pressure** (and also the chemical composition of the fluid if the fluid is not a pure substance)



1. Effect of Temperature:



Viscosity of liquid **decrease** with rising temp.(fluidity increase with temp.) because of the increasing Brownian motion → overcome the attraction forces among each other. Thus they are freed to move, and less resistance to flow is experienced...

- **Viscosity of gas increase with rising the temp.**
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2. Effect of Molecular size, shape and Intermolecular

- forces influences the viscosity Molecular weight :



liquid with **large** and **irregularly** shaped molecular are generally known to be **viscous** as compared to **small** and **symmetric molecules**. Molecular collision between larger molecular molecules are not elastic i.e. involve loss of kinetic energy.

As a result ,intermolecular interactions are stronger and the molecular tend to stick to each other there by increasing the viscosity of the liquid.





3. Effect of Pressure :

An increase in pressure enhances the cohesive forces of interaction leading to an increase in the viscosity.



4. Effect of added substances also influence the viscosity

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- In general small quantities of non electrolytes namely sucrose, glycerin and alcohol when added to the water the solution exhibits **increased** viscosity .
 - Similarly polymers and other macromolecules enhance the viscosity of solvent such as water.
 - On the other hand, small amount of strong electrolytes decrease the viscosity as Alkali metals and ammonium ions
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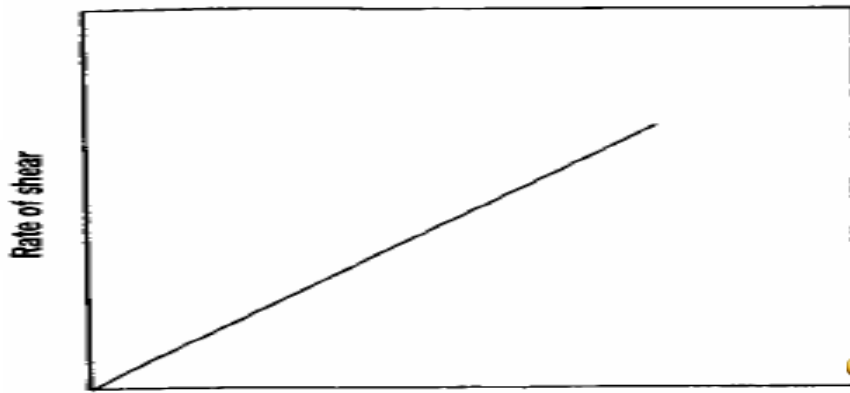
Non-Newtonian Flow

- **Non - Newtonian bodies** are those substances, which **fail** to follow Newton's law
- For Non- Newtonian fluids, the **viscosity** is affected by the **rate** of shear, or rate of shear history.
- Non-Newtonians do not show a fixed viscosity upon experiencing different shearing rates.
- Many polymer solutions, heterogeneous dispersions such as colloidal solutions, emulsions, liquid suspensions and ointments.

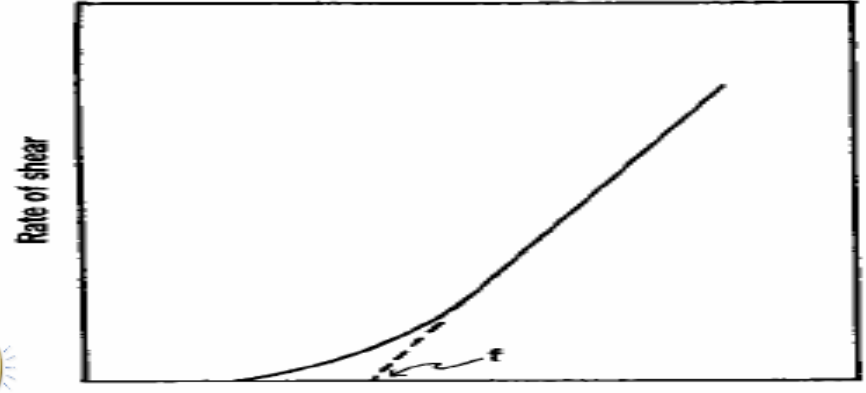
They are classified into 3 types of flow:

1. Plastic.
 2. Pseudoplastic.
 3. Dilatant.
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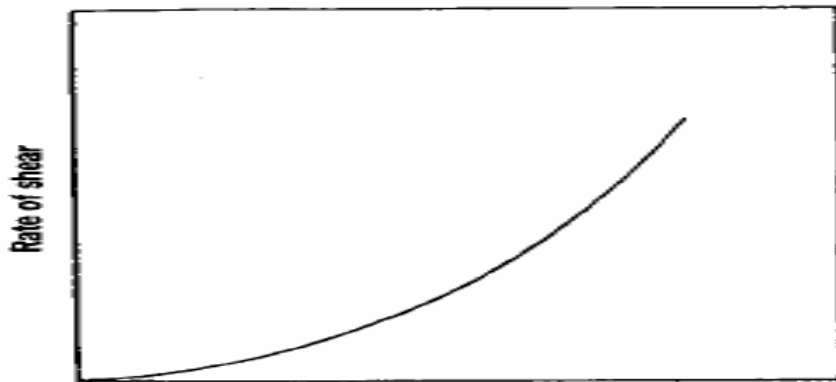
Rheograms of different fluids



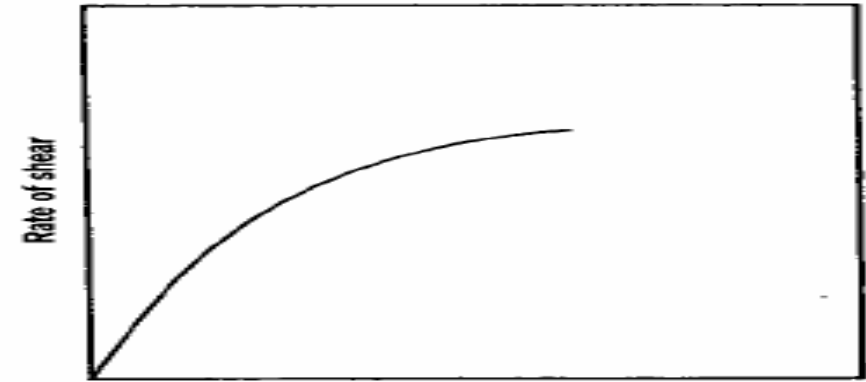
(a) Newtonian flow



(b) Simple plastic flow



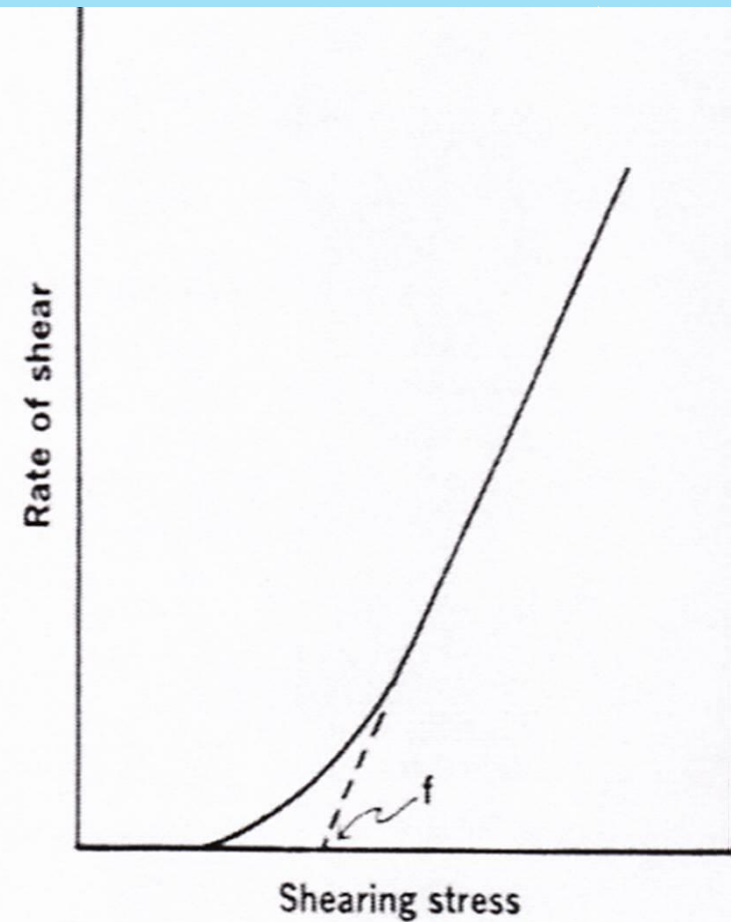
(c) Simple pseudoplastic flow



(d) Dilatant flow

1- Plastic Flow

- The plastic flow curve does not pass through the origin & it intersects the shearing stress axis (or will if the straight part of the curve is extrapolated to the axis) at a particular point referred to as **yield value. (f)**.
- The substances that exhibit plastic flow is called **Bingham bodies**
 - **Bingham bodies** doesn't begin to flow until a shearing stress exceeds the yield value
 - If stress is less than the yield value, the system behaves like a solid.

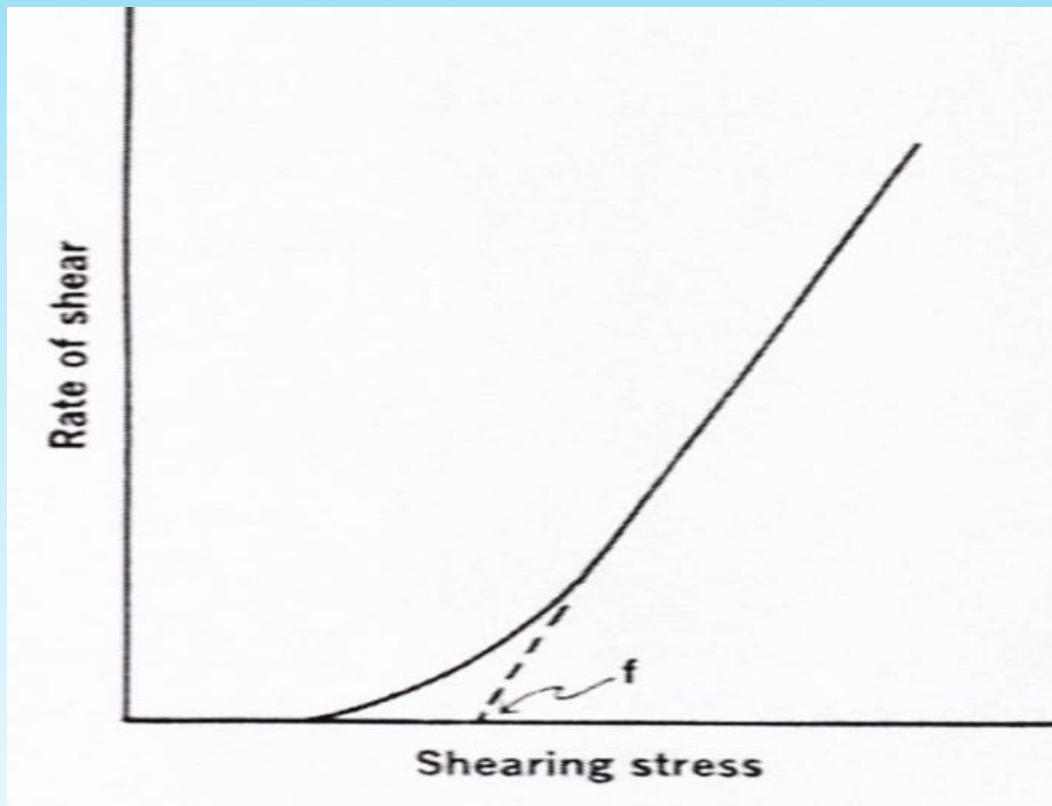


The material needs certain shear stress so as to start to flow ex: flocculated suspension

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- The viscosity **decreases** initially and then remains **constant**.
 - In a **highly flocculated system**, there is **interaction** between flocs which results in a structured system and plastic flow is associated with these systems e.g highly flocculated suspensions
 - **The yield value** is present because of the contacts between adjacent particles (caused by van derWaals forces which may be capable of withstanding weak stresses) also the friction forces between moving particles which must be broken down before flow can occur.
 - Consequently, the **yield value** is an indication of the **degree of flocculation**; the **more** flocculated the suspension, the **higher** will be the yield value.

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- Once the yield value has been exceeded, any increase in shearing stress (i.e., $F-f$) brings about a directly proportional increase in G , the rate of shear. In fact, a plastic system resembles a Newtonian system at shear stresses higher than the yield value.

- *A common example is **toothpaste**, which will not be extruded until a certain pressure is applied to the tube. It is then pushed out as a solid plug.*
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The slope of the rheogram → mobility (analogous to fluidity in Newtonian systems) and its reciprocal → Plastic viscosity, U .

$$U = \frac{(F - f)}{G}; \text{ as } f \text{ is the yield value.}$$